

Rewrite each series as a sum.

$$15) \sum_{n=1}^4 (20 - n^2)$$

$$16) \sum_{n=1}^4 (n + 600)$$

$$17) \sum_{m=1}^4 \frac{1}{3^m}$$

$$18) \sum_{a=1}^6 \frac{a^2 + 1}{a}$$

Evaluate each series.

$$19) \sum_{m=5}^{11} (m + 400)$$

$$20) \sum_{n=1}^5 \frac{10}{n}$$

$$21) \sum_{k=1}^6 \frac{300}{k}$$

$$22) \sum_{n=0}^5 (20 - n)$$

$$23) \sum_{k=1}^5 (200 - k^2)$$

$$24) \sum_{n=2}^8 n$$

$$25) \sum_{n=3}^9 \frac{1}{n}$$

$$26) \sum_{a=1}^7 \frac{a}{a+1}$$

Rewrite each series using sigma notation.

$$27) 1 + 4 + 9 + 16 + 25$$

$$28) 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5}$$

$$29) \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \frac{1}{81} + \frac{1}{243} + \frac{1}{729}$$

$$30) 5 + 10 + 15 + 20 + 25$$

$$31) \frac{1}{2} + \frac{2}{3} + \frac{3}{4} + \frac{4}{5} + \frac{5}{6}$$

$$32) 5 + 25 + 125 + 625 + 3125$$